



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

- Announcement
- Recap
- M1C3 begins

M1C3

One variable → Many variables →
Engineering equations → Uncertainty
propagation → Approximate propagation
→ Dependence beyond correlation

M1C3 lec-1

Joint Random Variables: Describing Multiple Uncertainties

How to describe uncertainty when two variables exist simultaneously?

Single RV → Joint Distribution → Marginals →
Conditional distributions → Covariance →
Correlation → Correlation $\neq$ Independence

Message: from uncertainty in **one variable** to
uncertainty in **systems**

L18

Sep 3, 2026

Messages from the Last Class

- **Message 1:** Joint random variables

Complete info of individual rv's is INSUFFICIENT

eg. ① Mutual funds A & B. Ind. prob. of losing = 10%.

$Pr(\text{Both will lose})$

could be $\frac{1}{2}$ 1% & 10%.

→ we need Joint Behavior.

② Building Damage: Demand D, Capacity C

$$Pr(\text{failure}) = Pr(D > C)$$

Requires
Joint Dist. of D & C.

Messages from the Last Class

- **Message 2: Joint PMF/PDF/CDF**

Joint CDF: $F_{XY}(x, y) = P_X(X \leq x, Y \leq y)$

Joint PMF: $P_{XY}(x_i, y_j) = P_X(X = x_i, Y = y_j)$

Joint PDF: $f_{XY}(x, y)$

$$F_{XY}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{XY}(u, v) du dv$$

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

$$F_{XY}(x, y) = \sum_{\substack{x_i \leq x, \\ y_j \leq y}} P_{XY}(x_i, y_j)$$

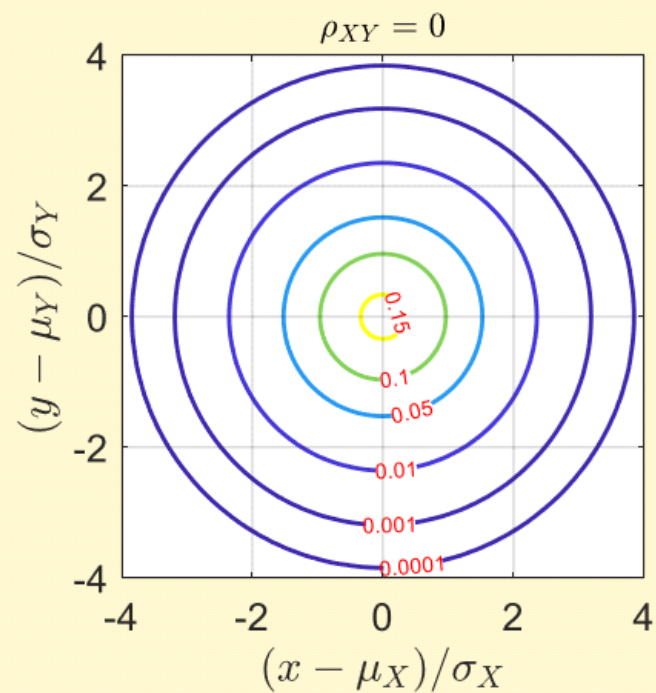
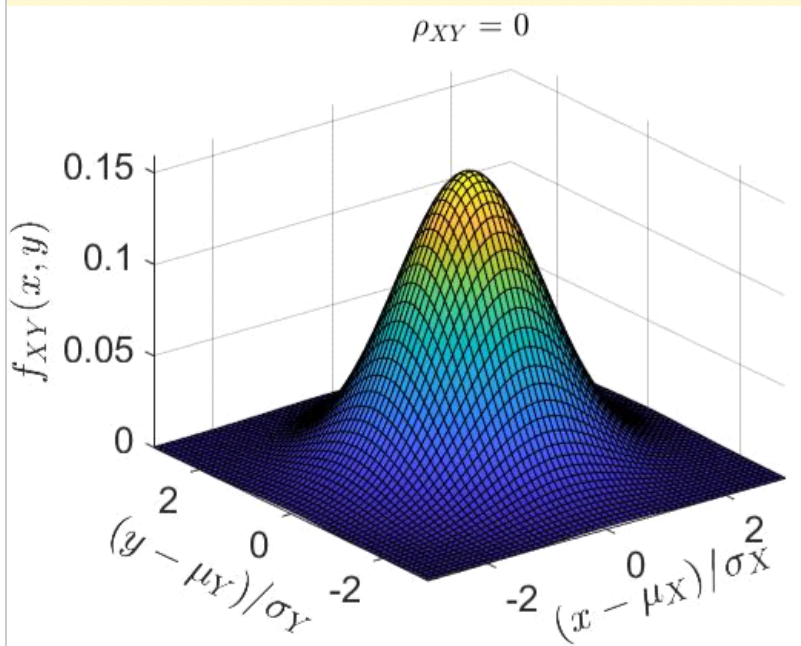
- **Message 3: Marginal probability distributions**

Joint Distribution $\longrightarrow$ Marginal Distribution

$P_X(x_i)$: Sum all $P_{XY}(x_i, y_j)$ over all y_j for $X = x_i$.

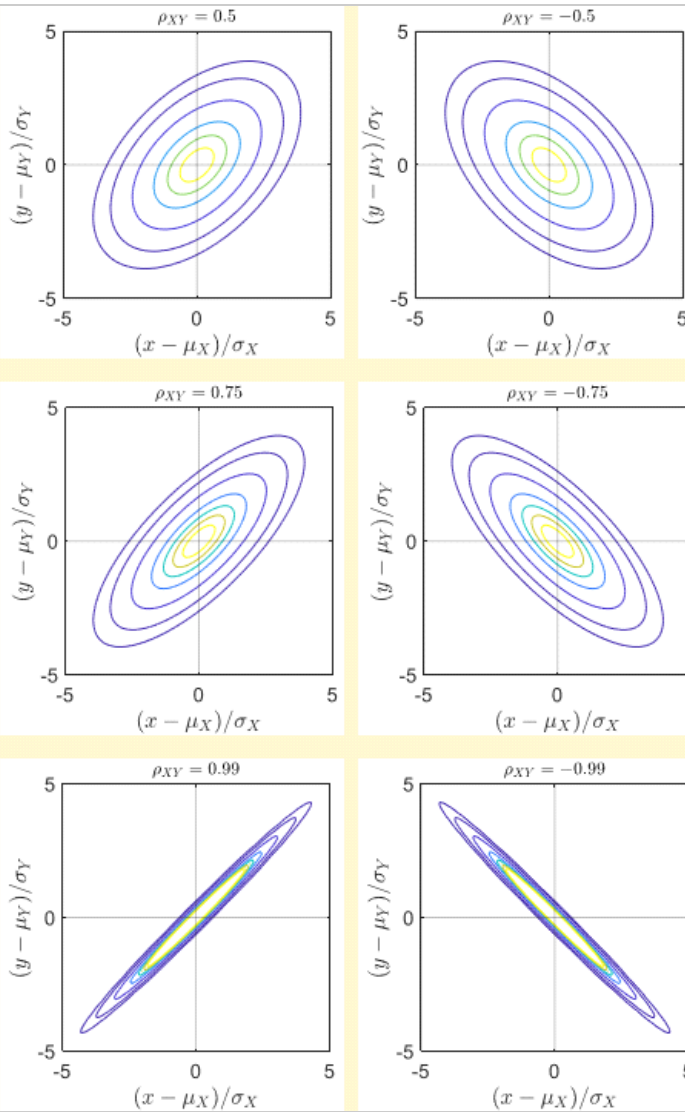
$f_X(x)$: Integrate $f_{XY}(x, y)$ over all y for $X = x$.

Joint PDF and marginals

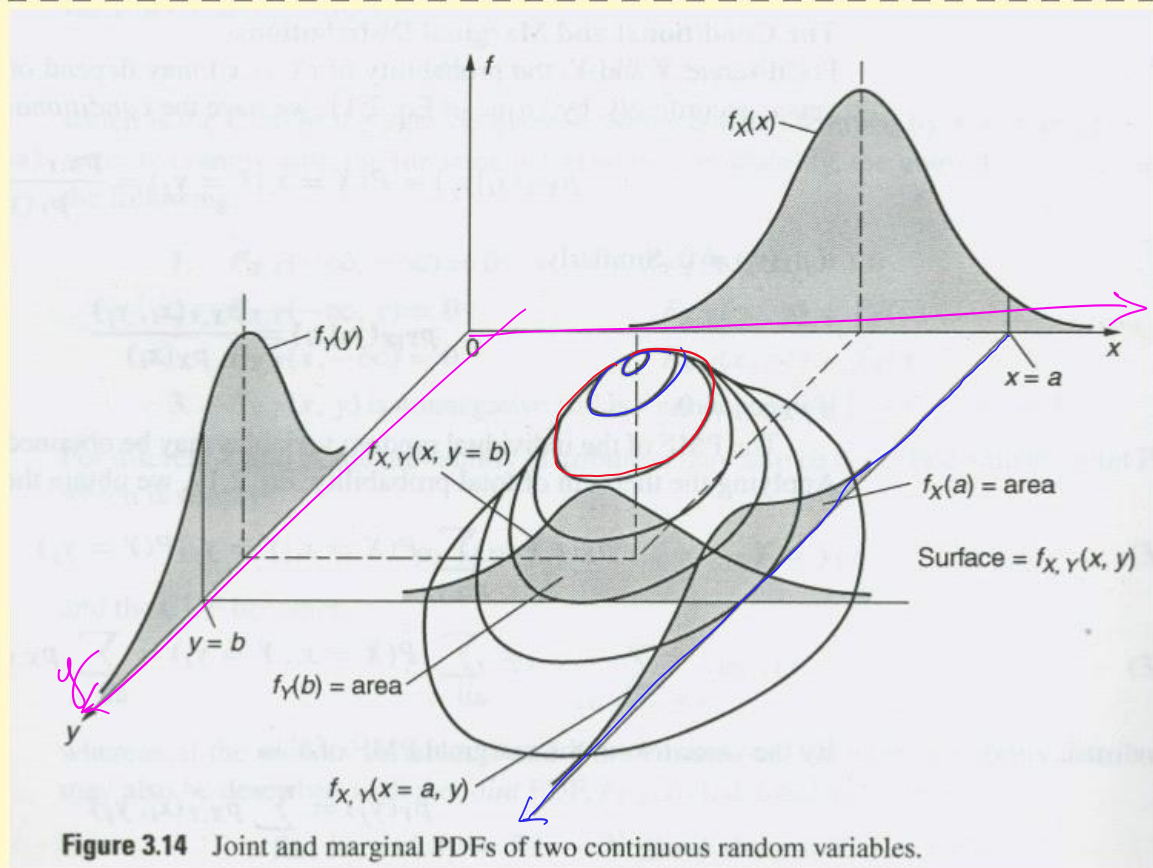


Through MATLAB program

Correlated Joint Distribution



Joint PDF and marginals



Joint probability distributions

$$1. \quad f_{xy}(x, y) = \frac{\partial^2 F_{xy}(x, y)}{\partial x \partial y}$$

$$② \quad F_{xy}(-\infty, -\infty) = 0$$

$$F_{xy}(-\infty, y) = 0$$

$$F_{xy}(x, \infty) = F_x(x)$$

$$F_{xy}(+\infty, +\infty) = 1$$

$$F_{xy}(x, -\infty) = 0$$

$$F_{xy}(x, \infty) = F_x(x)$$

$$③ \quad F_x(x) = \int_{-\infty}^{\infty} \int_{-\infty}^x f_{xy}(u, v) \, du \, dv$$

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, v) \, dv$$

$$= \frac{\partial F_x(x)}{\partial x}$$

Joint probability distributions

④ Joint PDF $\longrightarrow$ Marginals

$$f_{xy}(x, y) \neq f_x(x) f_y(y), \text{ In general}$$

Statistical Independence,

$$f_{xy}(x, y) = f_x(x) f_y(y)$$

⑤ Conditional probability

$$P_x(Y \leq y | X \leq x) = \frac{P_x(Y \leq y, X \leq x)}{P_x(X \leq x)}$$

Joint

Marginal

$$f_{Y|X}(y|x) = \frac{f_{xy}(x, y)}{f_x(x)}$$

$$\frac{\partial}{\partial y} F_{Y|X}(y|x) = \frac{\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} F_{xy} \right)}{\frac{\partial}{\partial x} (F_x)}$$

conditional CDF

Joint probability distributions

Joint probability distributions

Joint probability distributions

Joint probability distributions

Joint probability distributions

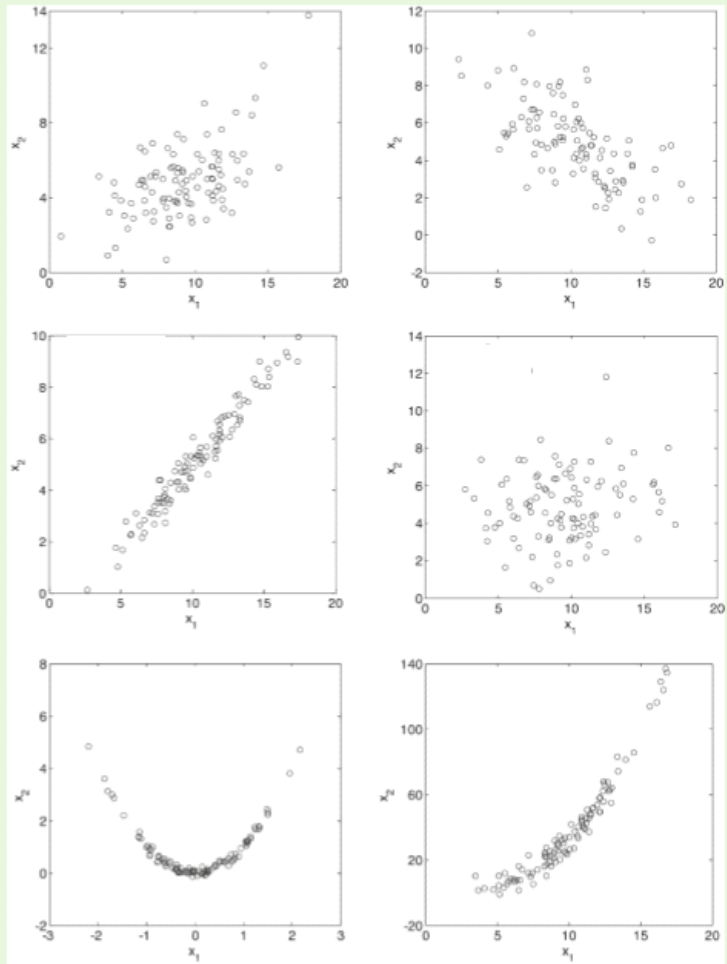
P14. Special case, Y is an affine transformation of X

$$Y = aX + b$$

$X: \mu_X, \sigma_X$ $Y: \mu_Y, \sigma_Y$

What is $\text{COV}[X, Y]$ and ρ_{XY} ?

Correlation coefficient



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Joint probability distributions

P14. Special case, Y is an affine transformation of X

$$Y = aX + b$$

$X: \mu_X, \sigma_X$ $Y: \mu_Y, \sigma_Y$

What is $\text{COV}[X, Y]$ and ρ_{XY} ?

Example on covariance

If X is either 11 or 17; Y is either 20 or 8.

Find $\text{COV}[X, Y]$ and ρ_{XY} .

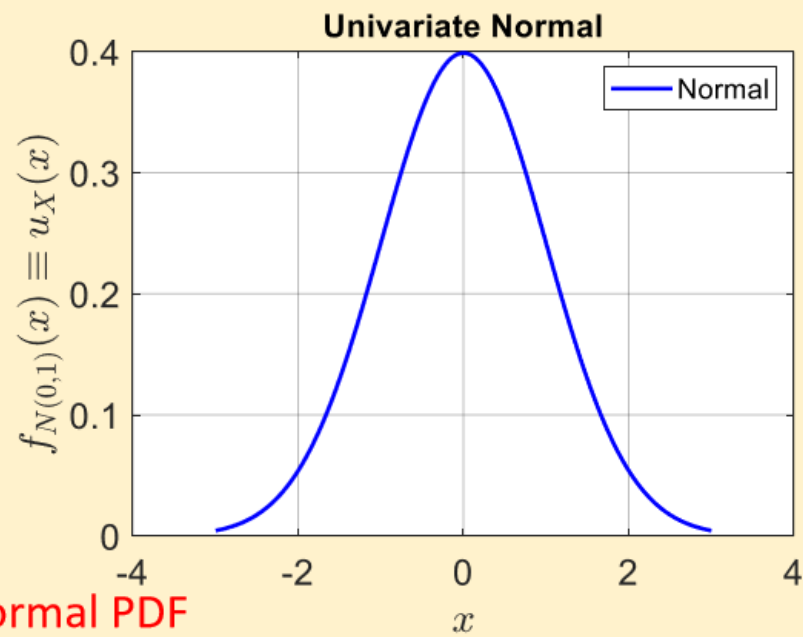
$$S = \{(11, 20), (11, 8), (17, 20), (17, 8)\}$$

(a) (11,20) and (17,8) have 0.5 probability each.

(b) All four with 0.25 probability each.

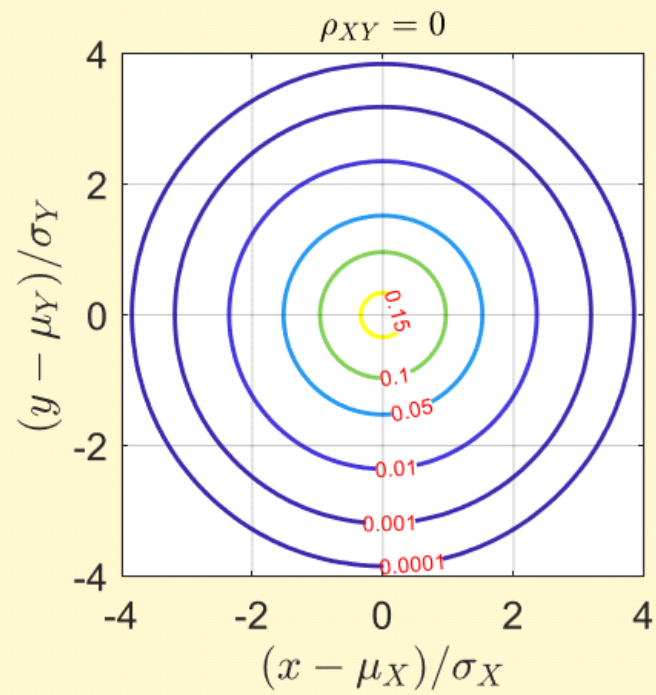
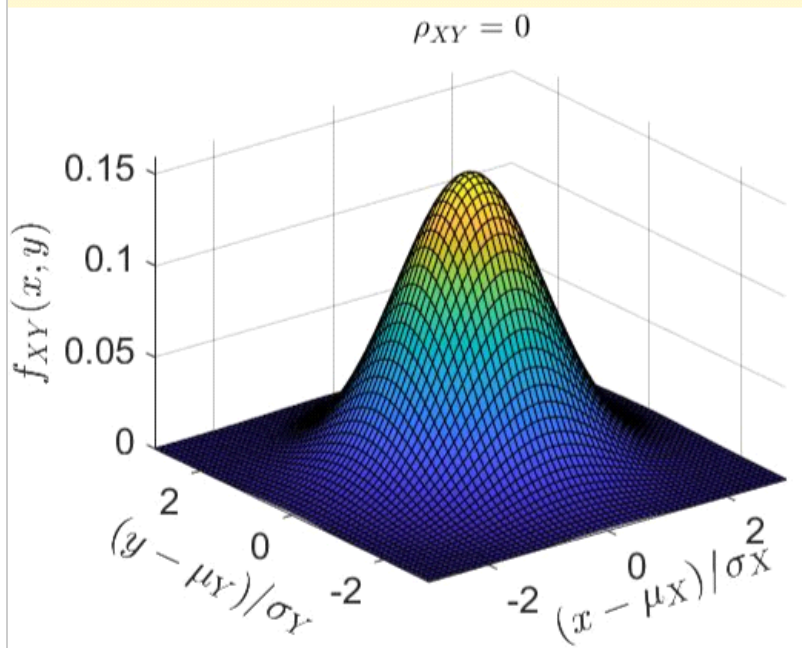
Multinormal distribution

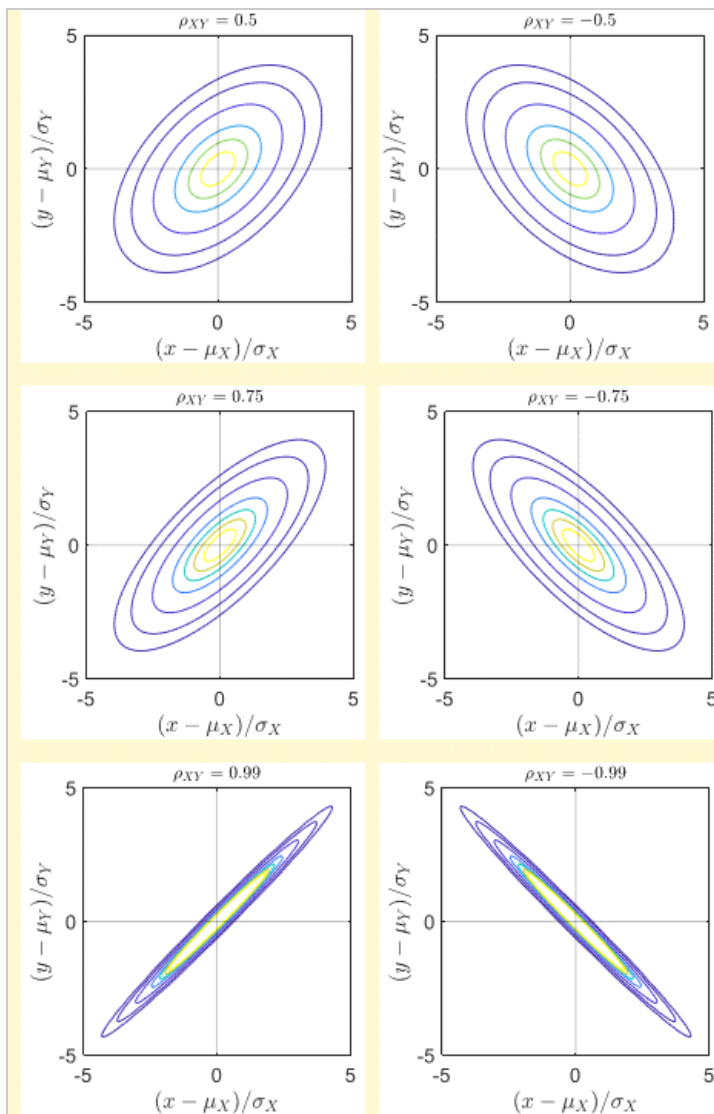
Univariate normal and standard normal



Univariate normal PDF

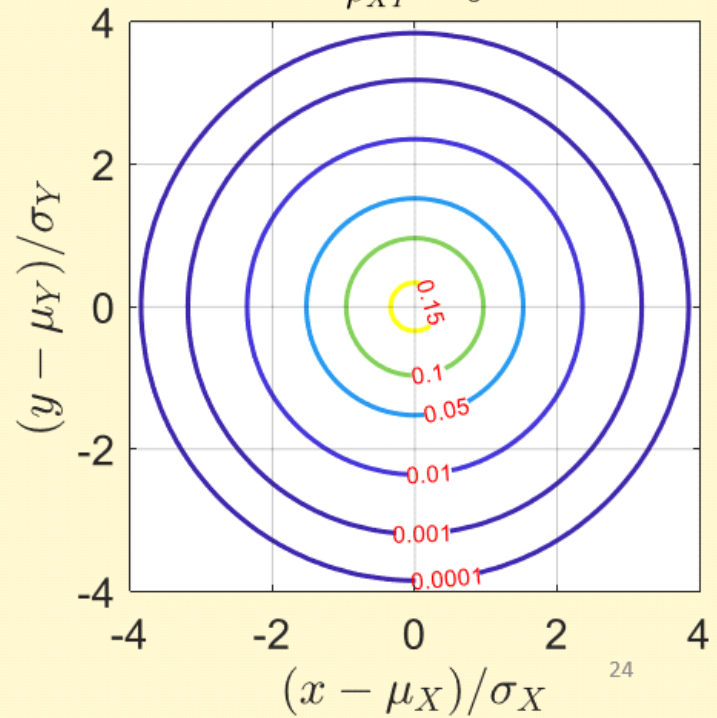
Bivariate normal and standard binormal





$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right] \right\}$$

$\rho_{XY} = 0$



Multinormal distribution: example

- X_1 and X_2 are jointly bivariate normal with mean and standard deviations $[\mu_1, \sigma_1]$ and $[\mu_2, \sigma_2]$, respectively. Their correlation coefficient is ρ . Find the distribution for $X_1|X_2$ (i.e., distribution type and distribution parameters).